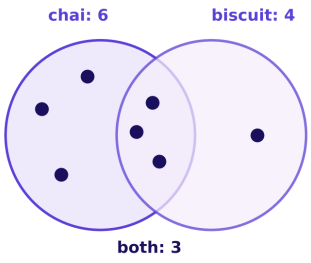
CHEAT SHEET

Which pairs beat chance, and what is a strip slot worth?

Swiggy Instamart, Pune, Week 4 Wednesday. The cart page's strip fell from 6.1 to 3.8 percent click-through once it showed the items most often found in the same cart. Illustrative case: the company is real, the situation and the data are synthetic, built for this class.

C2_W04_D03_beyond_chance_STUDENT
Verified 12 October 2026
7 panels, 1 diagram

1 Which three counts does every pair get?



ten orders, invented

	biscuit	no biscuit	all
chai	3	3	6
no chai	1	3	4
all	4	6	10

support 0.30 **confidence 0.50** **lift 1.25**

Count	The question it answers	Formula	On the ten dots
Support	How often are both in a cart?	both / all	3 / 10 = 0.30
Confidence	How reliably does A bring B?	both / A	3 / 6 = 0.50
Lift	How far above chance?	both x all / (A x B)	3 x 10 / (6 x 4) = 1.25

Crux. Lift is carts seen holding both over the carts chance gives, the same from either side; it counts and never names a cause.

2 What did twelve carts say, and what did 40,000 say?

Pair	Twelve carts	40,000 carts
bread, milk	confidence 1.00, lift 1.50	confidence 0.74, lift 1.04
chips, cold drink	lift 2.67 on 2 carts	lift 3.39 on 2,000 carts
ice cream, chocolate sauce	lift 12 on 1 cart	lift 4.60 on 100 carts

Crux. A few carts can make a pull look strong; milk is in 71 percent of carts, so its pull with anything sits near one.

3 Why did the click-through fall?

Slots showing	Share of impressions after	Click-through after
milk	50 percent, from 2 before	1.2 percent
everything else	50 percent	6.4 percent, from 6.2
blend	0.5 x 1.2 + 0.5 x 6.4	3.8 percent

Crux. The fall sits in the mix: milk took half the impressions, and 70 percent of slot 1, at 1.2 percent.

4 Which pairs fill the strip?

Region	Rule	Pairs
Promote	lift well above one, support at least 1 percent (400 carts)	chips and cold drink 3.39, coriander and green chillies 3.00, bread and butter 2.21
Watch	strong lift below the floor	ice cream and chocolate sauce, 4.60 on 100 carts
Ignore	lift near one	milk's pairs, 0.99 to 1.04

Crux. Rank by lift above a support floor; a strong pair on few carts is tried before it takes a slot.

5 What is a slot worth in rupees?

```
60,000 views a day x 6.4% x 0.5 adds per click x Rs 9 margin = Rs 17,280 a day
```

At 1.2 percent the slot earns Rs 3,240, a gap of Rs 14,040 a day or about Rs 4.2 lakh a month. Milk holds 1.5 slots' worth, so re-ranking recovers about Rs 6.3 lakh a month (Rs 3.9 lakh to Rs 9.3 lakh across the assumed ranges).

Crux. Value is margin on the adds, less any discount handed back; Rs 80,197 a day of sales is never the gain.

6 Which check catches each trap?

The trap	The check
Confidence 1.00, or a count ranking, read as a bundle	Lift: is the partner in most carts anyway?
"Shoppers changed"	Split by suggested item: the rest rose to 6.4
Heavy buyers inflate a pair	Count each customer once: 3.39 falls to 2.56
A stock-out hides a pair	Set it aside: curd's lift with milk stays 1.01
Lift read as a cause	Only a holdout that changes the strip settles it

Crux. Every count is asked "compared with what?" before it takes a slot.

7 What does the interviewer listen for?

"Lift, since it divides by the base rate: bread to milk reads 0.74, but milk is in 71 percent of carts, so lift is 1.04." An LLM shopping assistant's add-ons are scored by lift over a popularity baseline, then a holdout.

Crux. Name the base rate, the floor and the cost line, and end on one decision.

THE WORDS ON THIS SHEET

Support Share of all carts holding both items
Confidence Share of A's carts that also hold B
Lift Carts with both, over the carts chance alone gives
Base rate How common an item is across all carts
Support floor The fewest carts a pair needs to take a slot
Click-through Clicks divided by impressions